

ECO5050 AI Applications and Techniques in Economics

Lecture 5 Nonlinear Prediction Models and Factor Models

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Outline

- 1 Univariate Nonlinear Prediction Models
- 2 Principal Component Analysis
- 3 Factor Models
- 4 Dynamic Factor Models

Univariate Nonlinear Prediction Models

- It is likely that linear prediction models are misspecified for many economic variables
 - ▶ Recessions tend to be shorter than expansions, and economic recoveries can have very different dynamics from recessions (Pesaran and Potter 1997)
 - ▶ Large increase in oil prices have found to affect GDP growth differently from declines of a similar magnitude (Hamilton 1996)
 - ▶ Asset prices in crashes and recoveries (Brunnermeier, Nagel, and Pedersen 2008)

Univariate Nonlinear Prediction Models

- Consider nonlinear models that take the form

$$y_{t+1} = f(z_t) + \varepsilon_{t+1}$$

where $f(\cdot) \in \mathcal{F}^R$ is limited to sets of nonlinear models that are known up to a set of parameters

- Threshold autoregressive (TAR) model
- Smooth transition autoregressive (STAR) model
- Markov switching model

Threshold Autoregressive (TAR) Models

- Consider an extension of the $AR(p)$ model where the coefficients and the variance can vary across different ranges of y_{t-d} , for some $d \geq 0$

$$y_{t+1} = \begin{cases} \phi_{11}y_t + \cdots + \phi_{1p}y_{t-p+1} + \sigma_1\varepsilon_{t+1} & \text{if } y_{t-d} \in S_1 \\ \vdots & \vdots \\ \phi_{k1}y_t + \cdots + \phi_{kp}y_{t-p+1} + \sigma_k\varepsilon_{t+1} & \text{if } y_{t-d} \in S_k \end{cases} \quad \varepsilon_{t+1} \sim (0, 1)$$

- The states S_i cover all possible values for y_{t-d} , the endpoints of which are known as thresholds
- Also known as self-exciting TAR (SETAR) since the threshold depends on y_{t-d} as opposed to some exogenous variable

Threshold Autoregressive (TAR) Models

- When the thresholds $S_1, \dots, S_k$ and d are known, split the sample into k groups and parameter estimation is straightforward
- If the thresholds are unknown, the thresholds need to be estimated
 - ▶ Chan (1993) showed that the threshold value and the coefficients in each state can be consistently estimated when $k = 2$ and y_t is ergodic and strictly stationary
- The multistep forecast is a nonlinear functions of future values taken by y ; Monte Carlo simulation or the bootstrap is used to obtain the distribution of ε_t to generate multistep forecasts.

Smooth Transition Autoregressive (STAR) Models

- The STAR model is similar to the TAR model, the key difference being that it allows a smoother transition between the states.
- Example: the two-state TAR model

$$y_{t+1} = (\phi_{11}y_t + \cdots + \phi_{1p}y_{t-p+1} + \sigma_1\varepsilon_{t+1})\mathbf{1}(y_{t-d} \in S_1) \\ + (\phi_{21}y_t + \cdots + \phi_{2p}y_{t-p+1} + \sigma_2\varepsilon_{t+1})(1 - \mathbf{1}(y_{t-d} \in S_1))$$

if the indicator function $\mathbf{1}(\cdot)$ is replaced with a function $F(s_t; \gamma, c) \in (0, 1)$ that smoothly moves between 0 and 1, where s_t is data dependent, $\gamma > 0$ and c are parameters affecting transition across states

$$y_{t+1} = (\phi_{11}y_t + \cdots + \phi_{1p}y_{t-p+1} + \sigma_1\varepsilon_{t+1})F(s_t; \gamma, c) \\ + (\phi_{21}y_t + \cdots + \phi_{2p}y_{t-p+1} + \sigma_2\varepsilon_{t+1})(1 - F(s_t; \gamma, c))$$

Smooth Transition Autoregressive (STAR) Models

- The main STAR models
 - ▶ Logistic STAR (LSTAR): $F(s_t; \gamma, c) = (1 + \exp\{-\gamma(s_t - c)\})^{-1}$
 - ▶ Exponential STAR (ESTAR): $F(s_t; \gamma, c) = 1 - \exp\{-\gamma(s_t - c)^2\}$
- The LSTAR model approaches the TAR model when $\gamma \rightarrow \infty$, a linear AR model when $\gamma \rightarrow 0$.
- The ESTAR model does not nest the TAR model; it is close to TAR model in the second state when s_t is near c and is otherwise close to the TAR model in the first state.

Markov Switching Models

- Markov switching or regime switching models have become a popular class of models in economics and finance since their introduction into economics by Hamilton (1989).
 - ▶ They share similarities with TAR and STAR; the model introduce a number of regimes linked by a function that determines which regime occurs at a particular point in time.
- Markov switching model assumes that the regime indicator is driven by an underlying Markov process

$$y_{t+1} = \mu_{s_{t+1}} + \sigma_{s_{t+1}} \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, 1), \quad s_{t+1} \in \{1, \dots, K\}$$
$$p_{ij} = \Pr(S_t = i | S_{t-1} = j)$$

Markov Switching Models

- For the model with two regimes ($K = 2$) we can write the transition probability $P = [p_{ij}]$

$$P = \begin{pmatrix} p_{11} & 1 - p_{22} \\ 1 - p_{11} & p_{22} \end{pmatrix}$$

The first-order homogenous Markov chain assumes that the transition probabilities do not depend on data.

- Estimation of regime switching models is typically based on either maximum likelihood or Bayesian methods and assumes that the likelihood is Gaussian within each regime.
- See Chapter 22 of Hamilton (1994).

Markov Switching Models

Empirical example: Two-state regime switching models

- Two-state regime switching model to four variables: inflation, stock returns, unemployment and interest rates

$$y_{t+1} = \mu_{s_{t+1}} + \rho_{s_{t+1}} y_t + \sigma_{s_{t+1}} \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, 1)$$

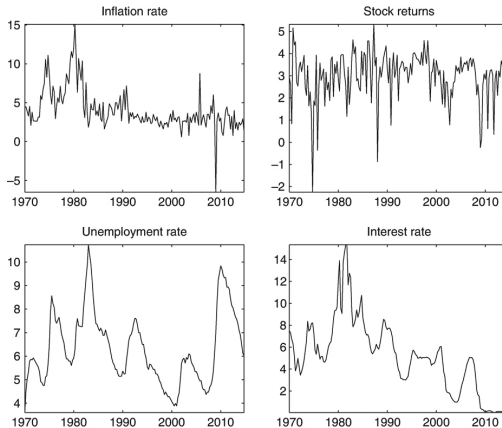
TABLE 8.1:

Maximum likelihood estimates of the parameters of two-state regime switching models: σ_1 and σ_2 measure the volatility parameters in states 1 and 2; μ_1 and μ_2 measure the intercept in states 1 and 2; ρ_1 and ρ_2 measure the autoregressive coefficient in the two states. Finally, p_{11} and p_{22} are the probabilities of remaining in states 1 and 2, respectively.

Parameters	Inflation rate	Stock returns	Unemployment rate	Interest rate
$\sigma_1(\%)$	4.9324	5.4268	0.1534	0.0871
$\sigma_2(\%)$	1.9049	10.3857	0.5772	0.8427
$\mu_1(\%)$	2.6561	4.0226	0.0901	0.0091
$\mu_2(\%)$	0.8200	1.1493	0.6442	0.1712
ρ_1	0.0946	0.0227	0.9667	1.0114
ρ_2	0.7897	0.0957	0.9302	0.9648
p_{11}	0.7549	0.9084	0.9385	0.8843
p_{22}	0.9633	0.8413	0.8797	0.9363

Markov Switching Models

Empirical example: Two-state regime switching models



- Recursive out-of-sample forecasts based on the two-state Markov switching models

Principal Component Analysis

- Principal Component Analysis (PCA) is a statistical technique that allows us to reduce the number of dimensions in the data while retaining as much as possible the variation present in the data.
- If we have 120 variables of real economic activity, do we need a 120 dimension space to view data? How do we find the best way to reduce the dimension space that conveys maximum useful information?
- PCA address this issue by ordering linear combination of the variables by their contribution to variance.

Principal Component Analysis

Let X be a $(k \times 1)$ random vector and $\Sigma = \text{Var}(X)$

- The first principal component is $U_1 = h_1'X$, where h_1 satisfies

$$h_1 = \arg \max_{h'h=1} \text{Var}(h'X) = \arg \max_{h'h=1} h'\Sigma h$$

- The second principal component is $U_2 = h_2'X$, where h_2 satisfies

$$h_2 = \arg \max_{h'h=1, h'h_1=0} \text{Var}(h'X)$$

In general, the j th principal component is $U_j = h_j'X$, where h_j satisfies

$$h_j = \arg \max_{h'h=1, h'h_1=0, \dots, h'h_{j-1}=0} \text{Var}(h'X)$$

Principal Component Analysis

- The principal components of X are linear combinations $h'X$ ranked by contribution to variance.
- Most of the variance in X will be captured by the first few principal components, and the variance of the rest of the principal components will be low so these can be components can be omitted from further analysis.
- Visual explanation: [▶ Link](#)

Principal Component Analysis

- The **eigenvalue** λ and associated **eigenvector** h of a $(k \times k)$ square matrix A satisfies $Ah = h\lambda$ where $h'h = 1$. Set $H = [h_1, \dots, h_k]$ and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_k)$ then

$$AH = H\Lambda$$

with $H'H = I_k$

- (Spectral Decomposition) If A is a $(k \times k)$ real symmetric matrix, then $A = H\Lambda H'$, or alternatively, $H'AH = \Lambda$

Principal Component Analysis

Theorem

The principal components of X are $U_j = h_j'X$, where h_j is the eigenvector of Σ associated with the j th ordered eigenvalue λ_j of Σ .

- Since Σ is symmetric the spectral decomposition states that $H'\Sigma H = \Lambda$ where $H = [h_1, \dots, h_k]$ and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_k)$.
 - ▶ Since Σ is positive semi-definite, the eigenvalues are real, nonnegative and ordered $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$
- Let $U = (U_1, \dots, U_k)$ be the principal components of X , by the Theorem, $U = H'X$ and

$$\text{Var}(U) = \text{Var}(H'X) = H'\Sigma H = \Lambda$$

which is diagonal. The principal components are mutually uncorrelated, $\text{Cov}(U_i, U_j) = 0$ for $i \neq j$.

- The principal components are the uncorrelated linear combinations of $X = (x_1, \dots, x_k)'$ whose variances are as large as possible.

Principal Component Analysis

- The relative variance component of the j th principal component is $\lambda_j / \text{tr}(\Sigma)$.
- Principal components are sensitive to the scaling of X ; it is recommended to first scale each element of X to have mean 0 and unit variance
- The sample principal components are obtained by replacing the unknowns by sample estimators
 - ▶ $\hat{\Sigma}$ is the sample covariance matrix, and $\hat{h}_1, \dots, \hat{h}_k$ its ordered eigenvectors. The sample principal components are $\hat{h}_j'X$

Factor Models

- Previously we learned model selection methods that can select a subset of predictor variables for a linear regression forecasting model.
- The factor model attempts to extract the salient information in X using dimensionality reduction techniques, which are particularly useful when the variables in X are collinear.
- A single **factor model (FM)** for individual x variables are linearly related to a common factor f_t is

$$x_{it} = \lambda_i f_t + u_{it}, \quad i = 1, \dots, N$$

where λ_i are factor loadings and u_{it} are idiosyncratic shocks

- $\{\lambda_i, f_t, u_{it}\}$ is unobserved, f_t and u_{it} are uncorrelated, $E(u_{it}) = 0$ and $E(u_{it}u_{js}) = 0$ for $i \neq j, t \neq s$
- The factor model specifies that the correlation between elements of X is due to the common factor f .
- Assume x_{it} are scaled to have mean 0 and unit variance.

Factor Models

- A multiple FM has $q < N$ factors

$$x_{it} = \underset{(1 \times q)}{\lambda_i'} \underset{(q \times 1)}{f_t} + u_{it}$$

where $f_t = (f_{1t}, \dots, f_{qt})'$ is an $(q \times 1)$ vector of factors.

- Let $X_t = (x_{1t}, \dots, x_{Nt})'$ be a $(N \times 1)$

$$\underset{(N \times 1)}{X_t} = \underset{(N \times q)}{\Lambda} \underset{(q \times 1)}{f_t} + \underset{(N \times 1)}{U_t}$$

- The hope in using factor analysis for forecasting is that there exists a decomposition with q small enough that the f_t factors explain a large amount of variation in X_t , but with far fewer variables when the factors are used rather than the original data.

Forecasting with Factor Models

- With a large set of potentially relevant predictor variables x_{it} , $i = 1, \dots, N$, the simplest forecasting approach would be to consider a linear model

$$y_{t+1} = \alpha + \sum_{i=1}^N \beta_i(L) x_{it} + \gamma(L) y_t + \varepsilon_{t+1} \quad (1)$$

where $\beta_i(L) = \sum_{j=0}^{p_x} \beta_{ij} L^j$ is a lag polynomial which contains the regression coefficients for the i th variable, and $\gamma(L) = \sum_{j=0}^{p_y} \gamma_j L^j$ is a lag polynomial of order p_y

- Ignoring potential serial correlation in ε_{t+1} , the number of coefficient parameters $N_\beta = N \times p_x + p_y + 1$, and it is often the case that $N_\beta > T$

Forecasting with Factor Models

- Instead, assume that while a few (n_{x^*}) key predictor variables x^* directly affect y , most of the x variables affect y only through a set of q common components f_t , where $q \ll N$

$$y_{t+1} = \alpha + \beta_f(L)'f_t + \gamma(L)y_t + \phi(L)'x_t^* + \varepsilon_{t+1} \quad (2)$$

where f_t is a set of common factors extracted from the list of N original predictors, X_t

- The model has $p_f \times q + p_y + p_{x^*} \times n_{x^*} + 1$ parameters, which typically is much smaller than N_β .
 - ▶ Example: $p_y = p_x = p_f = p_{x^*} = 2$, $N = 200$, $q = 2$, and $n_{x^*} = 2$. Then $N_\beta = 203$ in (1) while the model (2) requires estimating only 15 parameters

Forecasting with Factor Models

- In practice, the common factors are unknown and so must be extracted from the data. Forecasting with factor models can be thought as a two-step process.
 - 1 Estimates of the common factors f_t are extracted
 - 2 The factor estimates, along with past values of the predicted variable and other relevant variables are used to select and estimate a forecasting model.
- Suppose a set of factor estimates $\hat{f}_t$ have been extracted from the data, then the forecasts take the form

$$\hat{y}_{t+1|t} = \hat{\alpha} + \sum_{j=1}^{p_f} \hat{\beta}'_j \hat{f}_{t+1-j} + \sum_{j=1}^{p_y} \hat{\gamma}_j y_{t+1-j} + \sum_{j=1}^{p_{x^*}} \hat{\phi}'_j x_{t+1-j}^*$$

Factor Models

- A multiple FM with $q < N$ factors

$$X_t = \Lambda f_t + U_t \quad (3)$$

- The error vector U_t is assumed to be mean 0, uncorrelated with f_t , and (under correct specification) mutually uncorrelated elements, $\Psi = E[U_t U_t']$ is a diagonal matrix.
- Factor loadings and factors are not separately identified; it is unwise to attribute meaning to the individual factor loading estimate.
 - ▶ If $\Lambda^* = \Lambda G$ and $f^* = G' f$ where G is $(q \times q)$ and orthonormal, the regression model is identical (any orthogonal rotation gives an equally valid representation)

Factor Models

- A multiple FM with $q < N$ factors: $X_t = \Lambda f_t + U_t$
- The factors f can either be treated as random or as a regressor; for now treat it as random and assume it has mean 0 and are normalized, $E[f_t \cdot f_t'] = I_q$
- The assumptions imply the correlation matrix $\Sigma_{XX} = E[X_t X_t']$ equals

$$\Sigma_{XX} = \Lambda \Lambda' + \Psi$$

- The parameters Λ and Ψ are often estimated by maximum likelihood; under joint normality of (f, U) , $X \sim N(0, \Lambda \Lambda' + \Psi)$
- The factors f can also be estimated:
 - ▶ Bartlett scoring
 - ▶ Regression scoring

Approximate Factor Models

- In many economic applications, N is large and it is also more reasonable to believe the FM is an approximation.
- Chamberlaine and Rothschild (1983) introduced the approximate FM, which is the same as (3) but relaxes the assumption on the idiosyncratic error U so that $\Psi = E[U_t U_t']$ is unrestricted.
 - ▶ The exact FM assumes the idiosyncratic error u of the factor model are mutually uncorrelated (Ψ is diagonal)

$$\begin{aligned} X_t &= \Lambda f_t + U_t \\ \Sigma_{XX} &= \Lambda \Sigma_{ff} \Lambda' + \Psi \end{aligned}$$

- Treat the factors as unknown parameters and simultaneously estimate the factors f and factor loadings Λ by least squares

Approximate Factor Models

- Let $(X_1, \dots, X_T)$ be a sample centered at sample means. $(\hat{f}_1, \dots, \hat{f}_T, \hat{\Lambda})$ is the joint minimizer of

$$\min_{f_1, \dots, f_T, \Lambda} \frac{1}{T} \sum_{t=1}^T (X_t - \Lambda f_t)' (X_t - \Lambda f_t) \text{ subject to } \Lambda' \Lambda = I_q \quad (4)$$

- Use the concentration argument: for fixed Λ , $X_t = \Lambda f_t + U_t$ is a set of N equations with q unknowns f_t . The LS solutions is

$$\hat{f}_t(\Lambda) = (\Lambda' \Lambda)^{-1} \Lambda' X_t$$

- Substitute this into the LS criterion (4), we obtain the concentrated LS criterion for Λ .

$$T^{-1} \sum_{t=1}^T (X_t - \Lambda f_t)' (X_t - \Lambda f_t) = \text{tr} \left[\hat{\Sigma}_{XX} - (\Lambda' \Lambda)^{-1} \Lambda' \hat{\Sigma}_{XX} \Lambda \right]$$

where $\hat{\Sigma}_{XX} = T^{-1} \sum_{t=1}^T X_t X_t'$

Approximate Factor Models

Theorem

The LS estimator of the FM (3) under the normalization $\Lambda'\Lambda = I_q$ has the following solution:

- 1 Let $\hat{D} = \text{diag}(\hat{d}_1, \dots, \hat{d}_q)$ and $\hat{H} = (\hat{h}_1, \dots, \hat{h}_q)$ be the first q eigenvalues and eigenvectors of the sample covariance matrix $\hat{\Sigma}$.
- 2 $\hat{\Lambda} = \hat{H}$
- 3 $\hat{f}_t = \hat{\Lambda}'X_t = \hat{H}'X_t$

- The factor estimates are the first q principal components of X_t
- $T^{-1} \sum_{t=1}^T \hat{f}_t \hat{f}_t' = \hat{D}$
- The estimators $\{\hat{f}_t\}$ could be rescaled so that $T^{-1} \sum_{t=1}^T \hat{f}_t \hat{f}_t' = I_q$, however this is unnecessary if the only purpose is forecasting.

Dynamic Factor Models

- The **Dynamic Factor Model (DFM)** expresses observed time series as a distributed lag of unobserved factors plus an idiosyncratic disturbance, where both the factors and disturbances are in general serially correlated.

$$\begin{aligned}X_t &= \Lambda(L)f_t + u_t \\ \Gamma(L)f_t &= v_t \\ B(L)u_t &= e_t\end{aligned}\tag{5}$$

where $\Lambda(L)$, $A(L)$ and $B(L)$ are lag polynomials with p , p_f and p_u lags, respectively.

- Assume f_t and u_t are uncorrelated at all leads and lags, $E[f_t u'_s] = \mathbf{0}$ for all t and s .
- The premise of the DFM is that the covariation among economic time series variables at lags and leads can be traced to a few underlying unobserved series (factors).
 - ▶ The disturbances to these factors might represent the major aggregate shocks to the economy, such as demand or supply shocks.

Dynamic Factor Models

- Exact DFM: The idiosyncratic disturbances are uncorrelated across series, $E[u_{it}u_{js}] = 0$ for all t and s with $i \neq j$.
- Approximate DFM: Allows correlation of idiosyncratic disturbance across series.
- The static (stacked) form of the DFM:

$$X_t = \bar{\Lambda}F_t + u_t$$

$F_t = (f'_t, f'_{t-1}, \dots, f'_{t-p+1})$ is a $(r \times 1)$ vector with $r \leq pq$

$\bar{\Lambda} = (\lambda_0, \lambda_1, \dots, \lambda_{p-1})$ where λ_h is the $(N \times q)$ matrix of coefficients of the h th lag in $\Lambda(L)$

Dynamic Factor Models

- The variable to be forecasted, y_t , can be handled in two ways
 - 1 Include y_t in the X_t vector and model it as part of the system.
 - 2 Treat the forecasting equation for y_t as a single equation.
- The single forecasting equation: the DFM is assumed to carry over to y_t

$$y_t = \lambda_Y(L)' f_t + u_{Yt}$$
$$\delta_Y(L) u_{Yt} = v_{Yt}$$

where v_{Yt} is assumed to be unpredictable given current information. Then

$$y_{t+1} = \beta(L) f_t + \gamma(L)' y_t + \varepsilon_{t+1}$$

where $\beta(L) f_t = E[\delta_Y(L) \lambda_Y(L)' f_{t+1} | f_t, f_{t-1}, \dots]$, $\gamma(L) = L^{-1}(1 - \delta_Y(L))$.

Dynamic Factor Models

Estimation of Factors

- MLE: A three-step parametric estimation approach to obtain a exact DFM
 - ① Use the Kalman filter to estimate the parameters of a Gaussian likelihood function for $\{y_{t+1}, x_t\}$
 - ② The Kalman filter and smoother is used to extract the estimates of $\{f_t\}_{t=1}^T$
 - ③ The forecast is computed by projecting y_{t+1} on $\hat{f}_t$ and any other relevant observable variables
- Principal component (PC) estimation
 - ▶ Connor and Korajczyk (1986) first showed the consistency of the PC estimator of f_t for T fixed and $N \rightarrow \infty$ in the exact FM
 - ▶ Stock and Watson (2002a) proved consistency of the PC estimator under the approximate FM
 - ▶ Bai and Ng (2006) show that as $N \rightarrow \infty$, $T \rightarrow \infty$ and $N^2/T \rightarrow \infty$, the estimation error of the PC estimators can be ignored when they are used as regressors (they can be treated as data)

Dynamic Factor Models

Determining the number of common factors

- We need to consider how many factors to include in the model
 - ① Number of factors that enter the DFM: $X_t = \Lambda F_t + u_t$
 - ② Number of factors in the forecasting equation of y_{t+1}
- (1) Bai and Ng (2002) provide that $r = \dim(F_t)$ can be estimated consistently for approximate DFMs that can be written in static form.
 - ▶ $\hat{F}_k$ is the vector of k factors obtained through principal components and $V(\hat{\Lambda}_k, \hat{F}_k)$ is the SSR obtained from (4)
 - ▶ Three information criteria

$$IC_{p1}(k) = \ln(V(\hat{\Lambda}_k, \hat{F}_k)) + k \left(\frac{N+T}{NT} \right) \ln \left(\frac{NT}{N+T} \right)$$

$$IC_{p2}(k) = \ln(V(\hat{\Lambda}_k, \hat{F}_k)) + k \left(\frac{N+T}{NT} \right) \ln(\min(N, T))$$

$$IC_{p3}(k) = \ln(V(\hat{\Lambda}_k, \hat{F}_k)) + k \frac{(\min(N, T))}{\min(N, T)}$$

Dynamic Factor Models

Determining the number of common factors

- We need to consider how many factors to include in the model
 - ① Number of factors that enter the DFM: $X_t = \Lambda F_t + u_t$
 - ② Number of factors in the forecasting equation of y_{t+1}
- (1) Onatski (2010) provide an alternative consistent estimator of r which estimates r as the largest value of k for which the difference between eigenvalues k and $k + 1$ of $\hat{\Sigma}_{XX}$ exceed a threshold.
- (2) The number of factors in the y_{t+1} equation can be estimated using model selection methods.
 - ▶ Given an upper bound on the dimension and lags of F_t , Stock and Watson (2002b) uses information criteria

Dynamic Factor Models

Missing data and mixed data sampling frequencies

- When N is large, incomplete data and missing observations are important practical concerns for economic forecasters.
 - ▶ Date of the final observations on different series can differ because of timing of date releases
 - ▶ Different sampling frequencies, e.g. monthly and quarterly
- The solution to the LS problem (4) in terms of the eigenvalues of $\hat{\Sigma}_{XX}$ holds when all NT observations are nonmissing (balanced panel data)
- Stock and Watson (2002b) propose using a expectation-maximization (EM) algorithm that fills in missing observations.

$$V(\Lambda, F) = \sum_{i=1}^N \sum_{t=1}^T S_{it} (X_{it} - \lambda_i' F_t)^2$$

λ_i is the i th row of $\bar{\Lambda}$ and $S_{it} = 1$ if an observation on X_{it} is available and 0 otherwise.

Dynamic Factor Models

Missing data and mixed data sampling frequencies

- Minimization of $V(\Lambda, F)$ requires iterative methods

$$V(\Lambda, F) = \sum_{i=1}^N \sum_{t=1}^T S_{it} (X_{it} - \lambda_i' F_t)^2$$

- The j th iteration of the algorithm is as follows (Stock and Watson, 2002b)
 - 1 $\hat{\lambda}_i^{(j-1)}$ and $\hat{F}_t^{(j-1)}$ are from the $(j-1)$ st iteration
 - 2 Fill out the missing values $\hat{X}_{it} = S_{it} X_{it} + (1 - S_{it}) \hat{\lambda}_i^{(j-1)'} \hat{F}_t^{(j-1)}$
 - 3 Using the rebalanced data set extract a new set of factor estimates $\hat{F}_t^{(j)}$ and update $\hat{\lambda}_i^{(j)}$
 - 4 The process continues until convergence

Dynamic Factor Models

DFMs for macroeconomic monitoring and forecasting

- Two classical applications of DMFs
 - ▶ Real-time macroeconomic monitoring: index construction, nowcasting
 - ▶ Forecasting
- Macroeconomic monitoring has several challenges including
 - ▶ Available data change daily or even within a day (“ragged edge”)
 - ▶ Number of data releases and series contained within those releases is vast
- The latent factor in the DFM summarizes the comovements of the observed variables. So in a DFM with a single factor, an estimate of the latent factor is a natural index of the movements of the relevant time series → index of indicators of economic activity

Dynamic Factor Models

DFMs for macroeconomic monitoring and forecasting: Indexes of economic activity

- Stock and Watson (1989,1991) Experimental Coincident Index (XCI)
 - ▶ Released monthly through NBER from May 1989 to Dec 2003
- Crone and Clayton-Matthews (2005) State Coincident Index
 - ▶ Monthly real activity indices for US states released by the Federal Reserve Bank of Philadelphia since 2005
- Aruoba, Diebold, and Scotti (2009) ADS index
 - ▶ Weekly index using mixed-frequency data, released in real time by the Federal Reserve Bank of Philadelphia
- Chicago Fed National Activity Index (CFNAI)
 - ▶ Released by the Federal Reserve Bank of Chicago since Jan 2001
 - ▶ Principal components estimate of the common factor in 85 real activity variables based on the real activity index constructed in Stock and Watson (1999)

Dynamic Factor Models

DFMs for macroeconomic monitoring and forecasting: Nowcasting

- Nowcasting is the prediction of the present, the very near future and the very recent past (Giannone, Reichlin, Small 2008)
- Relevant in economics because key statistics on the present state of the economy are available with a significant delay.
 - ▶ Official estimate of quarterly GDP in the US is published approximately one month after the end of the reference quarter
- The basic principal of nowcasting is the exploitation of the information which is published early and at higher frequencies than the target variable of interest.
 - ▶ e.g. On tracking GDP, look at personal consumption which is available at monthly frequency, or variables related to the production side such as industry output
- The New York Fed Staff Nowcast [▶ Link](#)

Dynamic Factor Models

DFMs for macroeconomic monitoring and forecasting: Nowcasting

- Giannone et al. (2008) adapts the factor model and uses a two-step approach to nowcast quarterly GDP.
 - ▶ X_t consists of monthly observed variables, the parameters are estimated from an OLS regression on principal components extracted from a balanced panel which is created by considering only the sample for which all observations are available.
 - ▶ The factors are re-estimated by applying the Kalman smoother on the entire data set.
- Bańbura and Modugno (2014) uses the EM algorithm to obtain MLE of the DFM on datasets with arbitrary pattern of missing data.

Dynamic Factor Models

DFMs for macroeconomic monitoring and forecasting: Nowcasting

- Empirical application in Bańbura, Giannone, Modugno, and Reichlin (2013)
- Establish whether daily and weekly variables contribute to the precision of the nowcast of quarterly GDP growth
 - ▶ Total 24 series of which only GDP is quarterly
- Nowcasts obtained by recursive estimation using expanding window, the first observations is fixed to the first business day of Jan 1983, the evaluations ample is 1995-2010.
- To assess how the accuracy of the nowcast improves as information of the current quarter accumulates, evaluate at different points within a quarter.
 - ▶ Four times a month (7, 14, 24, 28), starting in the first month of the current quarter up to the first month of the following quarter

Dynamic Factor Models

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Number	Name	Frequency	Publication Delay (in Days After)	Transformation	
			Reference Period)	Log	Diff
1	Real Gross Domestic Product	Quarterly	28	×	×
2	Industrial Production Index	Monthly	14	×	×
3	Purchasing Manager Index, Manufacturing	Monthly	3		×
4	Real Disposable Personal Income	Monthly	29	×	×
5	Unemployment Rate	Monthly	7		
6	Employment, Non-farm Payrolls	Monthly	7	×	×
7	Personal Consumption Expenditure	Monthly	29	×	×
8	Housing Starts	Monthly	19	×	×
9	New Residential Sales	Monthly	26	×	×
10	Manufacturers' New Orders, Durable Goods	Monthly	27	×	×
11	Producer Price Index, Finished Goods	Monthly	13	×	×
12	Consumer Price Index, All Urban Consumers	Monthly	14	×	×
13	Imports	Monthly	43	×	×
14	Exports	Monthly	43	×	×
15	Philadelphia Fed Survey, General Business Conditions	Monthly	-10		×
16	Retail and Food Services Sales	Monthly	14	×	×
17	Conference Board Consumer Confidence	Monthly	-5		×
18	Bloomberg Consumer Comfort Index	Weekly	4		×
19	Initial Jobless Claims	Weekly	4	×	×
20	S&P 500 Index	Daily	1	×	×
21	Crude Oil, West Texas Intermediate (WTI)	Daily	1	×	×
22	10-Year Treasury Constant Maturity Rate	Daily	1		×
23	3-Month Treasury Bill, Secondary Market Rate	Daily	1		×
24	Trade Weighted Exchange Index, Major Currencies	Daily	1		×

Note: The publication delays are based on the data releases in January 2011. Negative numbers for surveys mean that they are released before the reference month is over.

Dynamic Factor Models

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- Compare root MSE (RMSE) of 4 models
 - ▶ Benchmark: include GDP, monthly, weekly and daily data, estimated based on one factor following Bańbura and Modugno (2014)
 - ▶ Monthly: include GDP and monthly data, estimated based on one factor following Bańbura and Modugno (2014)
 - ▶ BDCD: small monthly factor model based on five indicators (real GDP, industrial production, real disposable income, retail sales and employment.
 - ▶ Bridge: estimate a bridge equation; estimate a separate regression of GDP on each of the 23 predictors, where the monthly predictors are aggregated to quarterly frequency.
- Also consider the results for the survey of professional forecasters (SPF) which are available in the middle second month of each quarter.

Dynamic Factor Models

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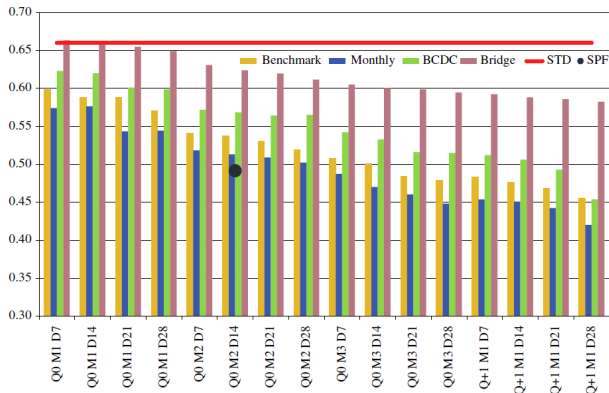


Figure 4.1 Root Mean Squared Forecast Error (RMSFE), GDP. *Notes:* Figure shows the Root Mean Squared Forecast Error (RMSFE) over period 1995–2010 for the model containing the complete dataset (Benchmark), the one that includes GDP and monthly variables only (Monthly), the one that uses the five hard variables considered by the NBER’s Business Cycle Dating Committee (BCDC) and the RMSFEs of the average forecast produced by the bridge equations (Bridge) over the sample period 1995–2010. The forecast accuracy of the model is evaluated four times per month for four consecutive months (from the first month of the current quarter, Q0 M1 to the first month of the following quarter, Q+1 M1). The date of the update is indicated on the horizontal axis. The dot corresponds to the RMSFE for the survey professional forecasters (SPF).